

Further Evidence for the Pool Function of Ubiquinone as Derived from the Inhibition of the Electron Transport by Antimycin

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The role of ubiquinone (Q) in the respiratory chain is studied by analyzing the inhibition by antimycin of the electron transport in uncoupled submitochondrial particles. As a result, the antimycin inhibition curve of substrate oxidation is quantitatively explained.

1. The participation of Q in the electron transport is required for the occurrence of nonlinear antimycin inhibition curves as demonstrated with external hydroquinones of different redox potentials as hydrogen donors. The respiration is non-linearly inhibited with a hydroquinone of sufficiently low redox potential to reduce the mitochondrial Q (menadiol) whereas a linear inhibition curve is found with hydroquinones of higher redox potentials (dihydroxybenzoquinone, dimethoxydimethylbenzohydroquinone). If Q is extracted, the inhibition curve also with menadiol becomes linear. This demonstrates the relation of the nonlinearity of the inhibition curve to the function of Q as a distributor of hydrogen between parallel cytochrome chains.

2. The electron transport through the cytochrome segment is inhibited linearly with increasing amounts of antimycin as demonstrated by the linear titration of three different functions: (a) the oxidation rate of Q_{red} on addition of oxygen to anaerobic submitochondrial particles, (b) the respiratory activity with added hydroquinones, (c) the acceptor activity as defined by the ratio of the respiration rate to the redox state of Q (Kröger, Klingenberg and Schweidler, 1973).

3. The ratio of donor to acceptor activity to Q (V_{red}/V_{ox}) is the main parameter in controlling the shape of the antimycin titration curve. This curve becomes more nonlinear with decreasing this ratio. Also the redox state of Q is nonlinearly dependent on the amount of antimycin and controlled by the donor/acceptor activity ratio.

4. A theoretical hyperbolic relation of the rate of substrate oxidation to the amount of antimycin is derived, involving the ratio donor to acceptor activity and the linear inhibition by antimycin of the electron transport through the cytochrome segment. With independently measured donor/acceptor activity ratios, antimycin titration curves are calculated which give a good fit to the experimental titrations over the range from highly nonlinear to almost linear curves. The observed nonlinearity is thus recognized to result from a hyperbolic relationship. Also the calculated titration curves for the redox state of Q are in agreement with the experiments.

5. The deviation of the experiment from the calculated curves near equimolarity between the amount of antimycin and inhibition sites can be accounted for by the dissociation of antimycin from the site of inhibition.

In a previous paper [1] the electron transfer rates to and from the Q-pool have been correlated to the respiratory activity. Thus quantitative support for the pool function of Q in the respiratory chain was gained. Another interesting insight into the function of Q as the hydrogen-transporting pool is offered by the nonlinear relation of the inhibition of substrate oxidation to the concentration of antimycin [2–6].

Abbreviations. Q, ubiquinone; Q_{ox} , oxidized ubiquinone; Q_{red} , reduced ubiquinone.

In this communication the nonlinear inhibition curve is explained by the electron-distributing function of Q. This is achieved on the basis of, firstly, the two-enzyme concept of the respiratory chain with Q as the interconnecting mediator as proposed in the previous paper [1] and, secondly, the linear dependence on antimycin of the inhibition of the electron transfer through the cytochrome segment of the respiratory chain [5]. The results also permit the dismissal of a recent theory [7–9] according to

which the "allosteric" interaction of antimycin is responsible for the nonlinear inhibition curve.

MATERIALS AND METHODS

Preparation of Submitochondrial Particles

Normal submitochondrial particles, submitochondrial particles deficient of cytochrome *c* and submitochondrial particles supplemented with cytochrome *c* were prepared from beef heart mitochondria as described previously [1]. Beef-heart mitochondria were prepared with bacterial proteinase according to A. L. Smith [10].

The concentration of the submitochondrial particles was referred to as protein content as determined by the modified biuret method with KCN [11].

Determination of the Respiratory Activity and of the Redox State of Q

The respiration was recorded using a micro Clark-Type oxygen electrode of our own design. The submitochondrial particles (0.5–2 mg/ml) were suspended in 0.7 ml standard incubation medium which consisted of 50 mM sodium/potassium phosphate pH 7.4. 1 μ mol *p*-trifluoromethoxycarbonyl cyanide phenylhydrazon/g protein was added throughout. Antimycin was added in ethanolic solution in quantities of 5 μ l using constriction pipettes. The concentration of the solution was measured photometrically at 320 nm ($\epsilon = 4.78 \text{ mM}^{-1} \cdot \text{cm}^{-1}$ [11a]). With the amounts of antimycin added the inhibition was mostly incomplete. With the following procedures the dehydrogenases were activated so that constant respiratory activities are obtained. With succinate (20 mM) as the substrate, the suspension was kept anaerobic after the exhaustion of oxygen by respiration. After the addition of antimycin, the respiration was initiated with catalase + H_2O_2 . With NADH as the substrate 50 μ M NADH was added after the addition of antimycin and allowed to be oxidized before the respiration was started again with 2 mM NADH. The respiratory activity with menadiol is accompanied by autooxidation even at pH 6.0. The rate of autooxidation was measured as the oxygen uptake at full respiratory inhibition with KCN.

In the steady state of respiration 0.5 ml of the suspension were sampled from the oxygen-electrode vessel and mixed with 2.5 ml of a mixture of methanol (60% v/v) and light petroleum. The extraction and determination of Q_{ox} and Q_{red} in these extracts was done as described earlier [1,11]. As expressed by Eqn (1) only part (Q_a) of the total Q (Q_t) is reducible even under equilibration conditions in the presence of KCN and substrate. Thus Q_t is divided up into an active (Q_a) and inactive part (Q_i).

$$Q_t = Q_a + Q_i. \quad (1)$$

In the following only the redox state of Q_a is considered. According to Eqn (2), Q_a consists of Q_{red} and Q_{ox} , both of which can vary between 0 and Q_a .

$$Q_a = Q_{\text{ox}} + Q_{\text{red}}. \quad (2)$$

Measurement of the Kinetics of Oxidation of Q_{red}

The kinetics of oxidation of Q_{red} were measured using either the moving-mixing chamber [12] with the dual-wavelength spectrophotometer at 280–289 nm [11,13] or the quench flow apparatus for the rapid extraction of Q. Both methods are described in the previous communication [1].

RESULTS

TITRATION OF THE ELECTRON TRANSPORT THROUGH THE CYTOCHROME CHAIN

In the preceding communication [1] the respiratory chain was treated as if it consists of a segment which reduces Q and the cytochrome chain which oxidizes Q. The interaction of antimycin with the electron transport is more directly studied by titrating the activity of electron flow through the cytochrome segment separately [5,14]. The rate of electron flow through the cytochrome segment (acceptor) from Q_{red} to oxygen was measured in submitochondrial particles by three independent methods [1], (a) as the respiratory rate with durohydroquinone and (b) as the rate of oxidation of Q_{red} . With a third method (c) the rate was calculated from the over-all respiratory activity and the redox state of Q_a in the steady-state, which was shown to reflect the acceptor activity V_{ox} for Q [1].

Titration of the Respiratory Activity with External Hydroquinones

The respiratory activity with certain hydroquinones is sensitive to antimycin [1,5,15,16,19,21] and independent of the presence of Q [1,5,21]. With menadiol, durohydroquinone, and 2,3-dimethoxy-5,6-dimethylbenzohydroquinone, three highly active electron donors for the respiratory chain are available, which differ considerably in their redox potential, ranging from $E'_0 = -1 \text{ mV}$ for menadiol [17], $E'_0 = 35 \text{ to } 60 \text{ mV}$, for durohydroquinone [17,18] to $E'_0 = 110 \text{ mV}$ for dimethoxy-dimethylbenzohydroquinone assuming the same E'_0 for dimethoxy-dimethylbenzohydroquinone as for Q [17]. Dependent on the donor used, the constituent Q can either participate or not in the electron transport, although it may not necessarily operate in the direct pathway. Thus it can be shown that the nonlinearity of the antimycin titration curve is correlated to the involvement of Q.

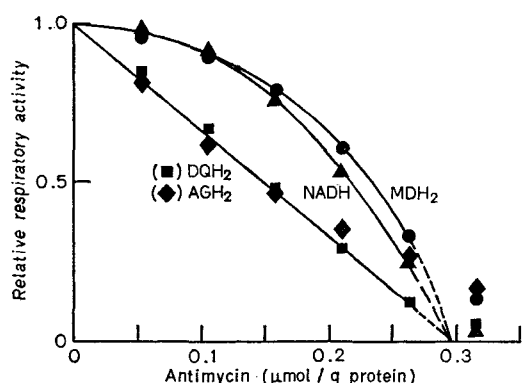


Fig. 1. Antimycin titration of the respiratory activity with various hydroquinones. The respiratory activity with durohydroquinone, dimethoxy-dimethylbenzohydroquinone, menadiol and NADH was measured in the presence of various amounts of antimycin. In each experiment antimycin was added stepwise to one single suspension of submitochondrial particles. Normal particles (0.14 mg/ml) were suspended in 0.1 M Tris pH 7.2 at 25 °C in the experiment with durohydroquinone, dimethoxy-dimethylbenzohydroquinone and NADH. The respiratory activity with NADH ($\blacktriangle$) was measured as described under Methods. The hydroquinones were generated by reduction of the quinones with NADH (2 mM) and D,T-diaphorase [20] in the presence of 2 μ mol/g protein rotenone. Experimental conditions: ($\blacksquare$), 0.83 μ M duroquinone and 0.3 mg/ml D,T-diaphorase. ($\blacklozenge$), 0.25 μ M dimethoxy-dimethylbenzoquinone and 0.8 mg/ml D,T-diaphorase. ($\bullet$), 0.14 mg/ml of the same preparation of particles were suspended in 40 mM phosphate buffer, 1 mM EDTA, pH 6.0 at 25 °C, 83 μ M menadion and 0.3 mg/ml D,T-diaphorase were added. The respiratory activities with durohydroquinone, dimethoxy-dimethylbenzohydroquinone, menadiol and NADH in the absence of antimycin were 350, 310, 300 and 453 μ atoms O/min per g protein respectively

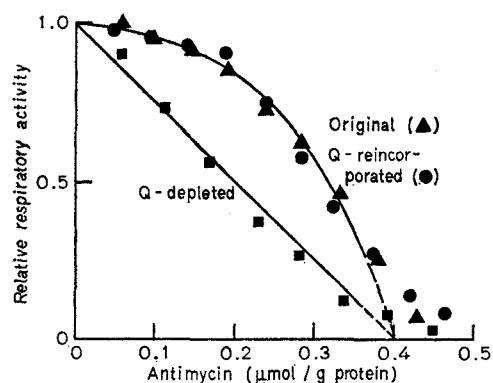


Fig. 2. Antimycin titration of the respiration with menadiol on depletion and reincorporation of Q. Depleted and reincorporated particles were prepared from normal submitochondrial particles according to Ernster *et al.* [21]. Each titration was done by stepwise addition of antimycin to one single sample of submitochondrial particles. The particles were suspended in 40 mM phosphate buffer, 1 mM EDTA, pH 6.0 at 25 °C with 2 μ mol/g protein rotenone, 2 mM NADH, 0.1 mg/ml D,T-diaphorase [20], 17 μ M menadione and 8 μ M cytochrome c. The respiratory activities of the original ($\blacktriangle$), depleted ($\blacksquare$) and reincorporated particles ($\bullet$) were 189, 137 and 98 μ atoms O/min per g protein, and the contents of Q were 6.3 (Q_t), 0.1 (Q_i) and 5.6 (Q_a) μ mol/g protein, respectively

In the experiment of Fig. 1 the respiratory activity with dimethoxy-dimethylbenzohydroquinone, durohydroquinone and menadiol, and for comparison with NADH, is titrated with antimycin. The hydroquinones are regenerated with D,T-diaphorase and NADH in the presence of rotenone [21]. The respiratory activity with the hydroquinones is adjusted by the amount of D,T-diaphorase and is lower than the respiratory activity with NADH. The titration curves are linear with durohydroquinone and dimethoxy-dimethylbenzohydroquinone and non-linear with menadiol and NADH. The same antimycin titer can be extrapolated from the titration curves obtained with the hydroquinones as with NADH, although the respiratory activities with menadiol and dimethoxy-dimethylbenzohydroquinone are not fully inhibited. This indicates a common pathway of the reducing equivalents from all the donors through the cytochrome chain.

The nonlinear shape of the antimycin titration curve with menadiol as the hydrogen donor can be explained by the interaction of Q. This is demonstrated by extraction and reincorporation of the Q of the particles (Fig. 2). On the extraction of Q the nonlinear curve is converted to a linear one and after the reincorporation of Q, again a nonlinear curve is obtained.

The relation of the nonlinearity to the interaction of Q can also be demonstrated by measuring the reduction of Q with menadiol and durohydroquinone as hydrogen donors. As shown by spectrophotometric recording in Fig. 3A, Q_a is fully reduced on the addition of 20 μ M menadiol in the presence of excess antimycin (upper trace), *i.e.*, the subsequent addition of succinate does not further increase the reduction. The artefact caused by 20 μ M menadiol is small, as shown by the final addition. Under the same conditions, the addition of 20 μ M durohydroquinone (centre trace) does not cause any reduction of Q. The slow absorption increase is due to the slow formation of duroquinone as a result of the incomplete inhibition by antimycin. The subsequent addition of succinate causes the total reduction of Q_a , as shown by comparison with the control experiment (lower trace), where succinate alone is added to the particles in the presence of antimycin. The experiment of Fig. 3B demonstrates that 70% of the *b*-cytochromes which absorb at 562–575 nm, are reduced by 8 μ M durohydroquinone in the presence of antimycin. When these experiments are repeated with KCN instead of antimycin, again Q is reduced only by menadiol and not by durohydroquinone (not shown).

Titration by Antimycin of the Rate of Oxidation of Q_{red}

In the experiment of Fig. 4, the initial rate of oxidation of Q_{red} is compared to the respiratory

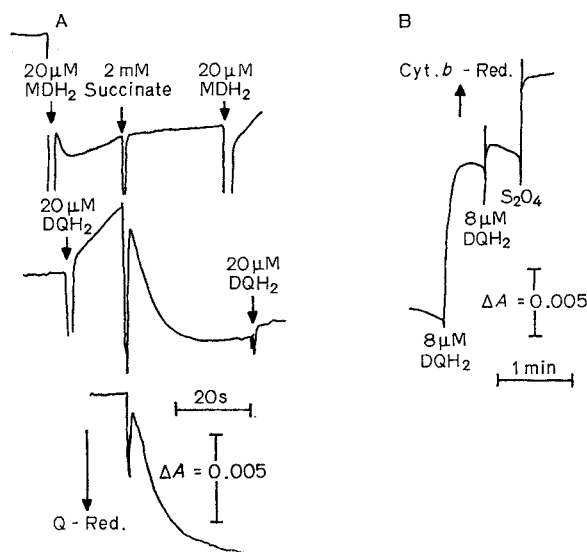


Fig. 3. Redox response of Q (A) and cytochrome b (B) to durohydroquinone and menadiol in the presence of antimycin. The redox response of Q was recorded at 280–289 nm [11, 13] and that of cytochrome b at 562–575 nm. Normal submitochondrial particles were suspended in 40 mM phosphate buffer, 1 mM EDTA, pH 6.0 at 22 °C and 0.7 μmol/g protein antimycin were added. A: the cuvette (optical path length 0.2 cm) contained 1.15 mg/ml particles. B: the cuvette (optical path length 0.5 cm) contained 2.15 mg/ml particles. The total content of Q was 6.3 μmol/g protein. MDH₂, menadiol; DQH₂, durohydroquinone

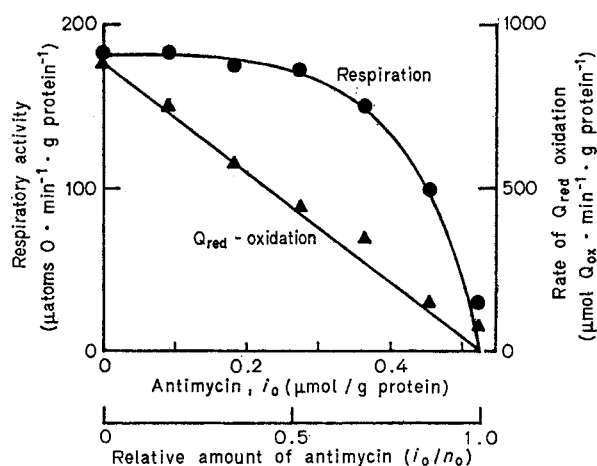


Fig. 4. Antimycin titration of the initial rate of Q_{red} -oxidation. Normal submitochondrial particles (2.5 mg/ml) were kept anaerobic at 15 °C in 0.1 M Tris, 2 mM EDTA, 20 mM succinate, 1 mM malonate, pH 7.2, with the concentrations of antimycin indicated on the abscissa. At each concentration of antimycin the respiratory activity was determined (●), and the initial rate of oxidation of Q_{red} was measured (▲) as the absorbance increase at 280–289 nm caused by the addition of oxygen-saturated buffer to the anaerobic suspension of the particles using the moving-mixing chamber [1, 12]. The second abscissa gives the amount of antimycin (i_0) relative to the antimycin-titer ($n_0 = 0.52$ μmol/g protein). Q_a was 6.0 μmol/g protein

activity in the presence of various amounts of antimycin (i_0). The oxidation of Q_{red} is measured spectrophotometrically after the rapid addition of oxygen to the anaerobic suspension with the moving-mixing chamber [12]. A striking difference in the dependence on antimycin of the two activities is observed: the respiratory activity is a nonlinear function, whereas the initial rate is a linear function of the amount of antimycin. Thus at an amount of antimycin causing 50% inhibition of the oxidation rate, almost no respiratory inhibition is observed. Both curves are extrapolated to cut the abscissa at the same antimycin titer ($n_0 = 0.52$ μmol/g protein). This indicates that antimycin inhibits both activities at the same site.

For further evaluation a second abscissa is introduced which gives the amount of antimycin relative to n_0 . Since the initial rate of oxidation of Q_{red} was demonstrated to be equal to the acceptor activity [1] in the absence of antimycin (V_{ox}), Eqn (3) holds, where V'_{ox} designates the acceptor activity in the presence of antimycin.

$$V'_{ox}/V_{ox} = 1 - i_0/n_0. \quad (3)$$

This equation is confirmed by the linear titration curves of the respiratory activity with external hydroquinones (Fig. 1 and 2) and by the experiment of Fig. 5.

It is reasonable to conclude from Eqn (3) that V'_{ox} is proportional to the amount of free inhibition sites n :

$$V'_{ox}/V_{ox} = n/n_0. \quad (4)$$

Therefore Eqn (3) becomes:

$$n/n_0 = 1 - i_0/n_0. \quad (5)$$

In Eqns (3) and (5) the dissociation of antimycin from the inhibition site is neglected. This is justified for a first approximation in view of the high affinity of the particles for antimycin [8].

Titration of the Redox State of Q by Antimycin

In the experiment of Fig. 5 the respiratory activity with NADH is decreased stepwise by antimycin and in each steady state the redox state of Q_a is measured. The respiratory activity v and the corresponding degree of reduction of Q_a are plotted against the relative amount of antimycin, i_0/n_0 . In addition the ratio of the respiratory activity to the degree of reduction ($v \cdot Q_a/Q_{red}$) is plotted. Both, v and Q_{red}/Q_a change nonlinearly with the concentration of antimycin, however, $v \cdot Q_a/Q_{red}$ is a linear function of i_0 . This is explained by Eqn (6) which relates the respiratory activity to the degree of reduction Q_{red}/Q_a and to the acceptor activity V'_{ox} .

$$v = V'_{ox} \cdot Q_{red}/Q_a. \quad (6)$$

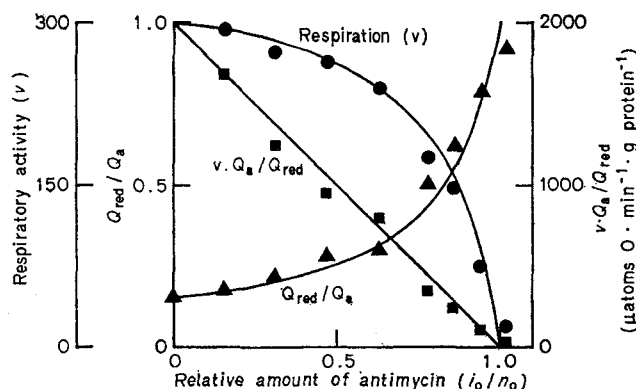


Fig. 5. The relation of the acceptor activity, as calculated from the respiratory activity and the corresponding degree of reduction of Q_a , to the amount of antimycin. Normal submitochondrial particles (1.6 mg/ml) were incubated in standard incubation medium at 25 °C with 5 μ M cytochrome *c* and 0.8 mM amytal at 25 °C. The respiratory activity with NADH, v , (●) and the degree of reduction of Q_a (Q_{red}/Q_a) (▲) were determined in the presence of various concentrations of antimycin as described under Methods. The abscissa gives the amount of antimycin (i_0) relative to n_0 (0.47 μ mol/g protein). Q_a and Q_t were 4.4 and 5.8 μ mol/g protein, respectively. (■) $v \cdot Q_a/Q_{red}$

According to Eqn (6) $v \cdot Q_a/Q_{red}$ is equal to V'_{ox} which was shown in Eqn (4) to obey a linear function of i_0/n_0 in agreement with the experiment of Fig. 5.

In the preceding communication [see Eqn (4) in [1]] v was found to be related to Q_{red}/Q_a and V_{ox} by a similar equation which was derived for the titration of the donor activity, when V_{ox} remains constant. The validity of this equation is extended here to the titration also of the acceptor activity.

In summary, the activity of electron transport through the cytochrome chain is demonstrated to be a linear function of the amount of antimycin. The nonlinearity in the titration curve of the respiration depends on the participation of Q .

THE ANTIMYCIN-TITRATION CURVE AS A FUNCTION OF THE DONOR AND THE ACCEPTOR ACTIVITY

In the preceding communication [1] it was shown that the respiratory activity, v_0 , in a given steady state can be described as a function of the donor and acceptor activity of Q only:

$$v_0 = V_{red} \cdot V_{ox} / (V_{red} + V_{ox}). \quad (7)$$

In the presence of various concentrations of antimycin, the analogous Eqn (8) holds:

$$v = V_{red} \cdot V'_{ox} / (V_{red} + V'_{ox}). \quad (8)$$

The substitution of V'_{ox} by Eqn (4) and the division of Eqn (8) by Eqn (7) results in Eqn (9) which

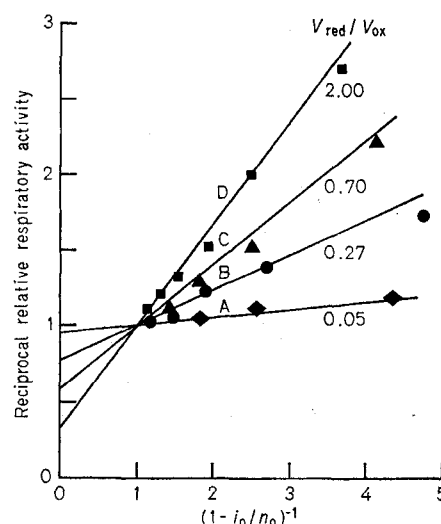


Fig. 6. Determination of V_{red}/V_{ox} by antimycin titration of the respiratory activity. The reciprocal relative respiratory activity (v_0/v) is plotted against $1/(1 - i_0/n_0)$ which is equal to n_0/n according to Eqn (5). The capital letters (A-D) refer to the four experiments given in Fig. 8 from which the data are taken. The straight lines are drawn for comparison according to Eqn (9) using the values of V_{red}/V_{ox} given in the figure (from Table 1)

relates the relative respiratory activity to the ratio donor to acceptor activity (V_{red}/V_{ox}) and n/n_0 .

$$v/v_0 = \frac{n/n_0 (1 + V_{red}/V_{ox})}{n/n_0 + V_{red}/V_{ox}} \quad (9)$$

or

$$v_0/v = \frac{1 + (V_{red}/V_{ox})n_0/n}{1 + V_{red}/V_{ox}}. \quad (9a)$$

According to Eqn (9) the relation of the respiratory activity to the amount of free inhibition sites is hyperbolic, since the reciprocal parameters [Eqn (9a)] are linearly related. A double reciprocal plot of the experimentally measured parameters is therefore expected to result in straight lines, the slopes of which should give the ratio V_{red}/V_{ox} . Since V_{red} and V_{ox} can also be measured by other methods, the validity of Eqn (9) for the antimycin titration can be tested.

In the experiment of Fig. 6 the respiratory activity with NADH and succinate is titrated with antimycin at four different ratios V_{red}/V_{ox} and the values for v_0/v are plotted against $1/(1 - i_0/n_0)$ which according to Eqn (5) is equal to n_0/n . The four straight lines of Fig. 6 are drawn according to Eqn (9a) with the given values of V_{red}/V_{ox} . V_{red} is varied by amytal and malonate and is determined from the respiratory activity and the corresponding redox state of Q_a in various states of antimycin inhibition as described in the previous paper [1]. V_{ox} is varied by changing the content of cytochrome *c* of the particles and is

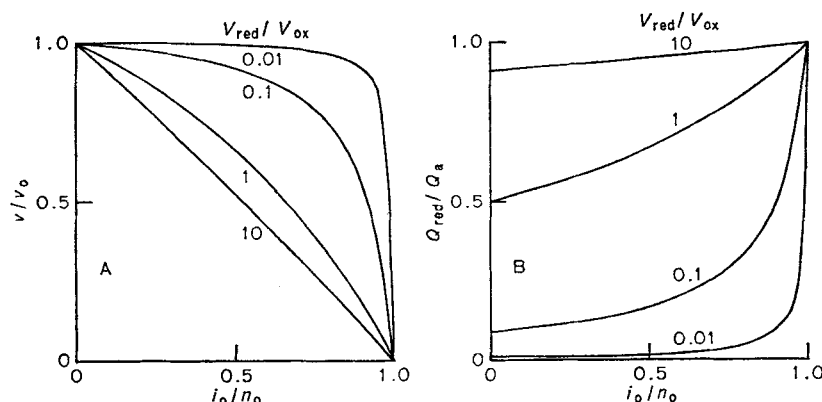


Fig. 7. Theoretical antimycin titration curves for the respiratory activity (A) and the degree of reduction of Q_a (B). The relative respiratory activity (v/v_0) and the degree of reduction of Q_a (Q_{red}/Q_a) with four different ratios V_{red}/V_{ox} are plotted against i_0/n_0 according to Eqn (9) and Eqn (11) respectively with n/n_0 given by Eqn (5)

Table 1. The kinetic constants required for the comparison of Eqns (9) and (11) to the experimental data of Fig. 8

The capital letters (A–D) refer to the same four experiments given in Fig. 6 and 8. The respiratory activity in the absence of antimycin (v_0) and the content of the redox active Q (Q_a) of the submitochondrial particles were determined as described under Methods. V_{red} is measured as the slope of the straight line obtained from the plot [1] of the respiratory activity against the corresponding degree of oxidation of Q_a in the steady state using the data of the experiment of Fig. 8. V_{ox} is measured as the initial rate of oxidation of Q_{red} on aeration of the anaerobic particles (see Methods and [1]). The experimental conditions are given in the legend of Fig. 8. The total content of Q (Q_t) was 5.8 (A), 6.3 (B), 4.1 (C), 5.0 (D) $\mu\text{mol/g protein}$

Expt	Substrate	v_0	V_{red}	V_{ox}	$\frac{V_{red}}{V_{ox}}$	n_0	Q_a
$\mu\text{atoms O} \times \text{min}^{-1} \times \text{g protein}^{-1}$							
A	NADH	97	101	2060	0.05	0.49	4.4
B	succinate	710	910	3410	0.27	0.60	5.4
C	succinate	350	638	910	0.70	0.56	3.8
D	NADH	450	1373	687	2.00	0.59	4.4

measured as the initial rate of oxidation of Q_{red} on aeration of the anaerobic particles [1]. These data are listed in Table 1 and are designated by the capital letters A–D referring to the four experiments given in Fig. 6. The good fit of the lines with the experimental points proves the validity of Eqn (9) for the antimycin titration and thus demonstrates that the respiratory activity is hyperbolically related to the amount of antimycin.

Also the degree of reduction of Q_a can be described as a function of the donor and the acceptor activity [1]. In the presence of antimycin the relation is given by Eqn (10) which on substitution of V'_{ox} by Eqn (4) gives Eqn (11).

$$Q_{red}/Q_a = V_{red}/(V_{red} + V'_{ox}) \quad (10)$$

$$Q_{red}/Q_a = \frac{V_{red}/V_{ox}}{n/n_0 + V_{red}/V_{ox}} \quad (11)$$

or

$$\frac{Q_a}{Q_{red}} = \frac{n/n_0}{V_{red}/V_{ox}} + 1. \quad (11a)$$

Although the reciprocal plots of Eqn (9a) and (11a) are more suitable for evaluating the ratio V_{red}/V_{ox} , there is an interest in the direct plots which illustrate the variation of the titration curve between strongly hyperbolic and almost linear. To illustrate Eqns (9) and (11), in Fig. 7 theoretical v/v_0 and Q_{red}/Q_a are plotted as functions of i_0/n_0 which is equal to $1 - n/n_0$ according to Eqn (5), with ratios V_{red}/V_{ox} varied 1000-fold. Extremely hyperbolic curves result from values of V_{red}/V_{ox} below 0.1, whereas with values greater than 1 the curves are almost straight.

The dependence of the shape of the experimental titration curve on V_{red}/V_{ox} is demonstrated in Fig. 8. The relative respiratory activity and the degree of reduction of Q_a in the corresponding steady state is plotted against i_0/n_0 . Both these parameters are compared to the theoretical curves drawn according to Eqns (9) and (11) with n/n_0 given by Eqn (5) and the given values of V_{red}/V_{ox} .

In agreement with the theoretical curves, the experimentally observed relations between the respiratory activity and the amount of antimycin depend on the ratio V_{red}/V_{ox} and vary from strongly hyperbolic (Fig. 8A) to almost straight (Fig. 8D). With the smallest value ($V_{red}/V_{ox} = 0.05$) the most hyperbolic relation is observed and with the greatest value ($V_{red}/V_{ox} = 2$) the relation is almost linear. The shape of the curves in Fig. 7B and C are intermediate in accordance with the corresponding ratios V_{red}/V_{ox} . Also the degree of reduction of Q_a responds as predicted by the model. This confirms that the total Q-pool represents a functional entity [1]. It can be excluded that the absolute values of V_{red} , V_{ox} or v_0 (Table 1) are responsible for the variation of the titration curve, since these values vary independently of V_{red}/V_{ox} and of the shape of the titration curve.

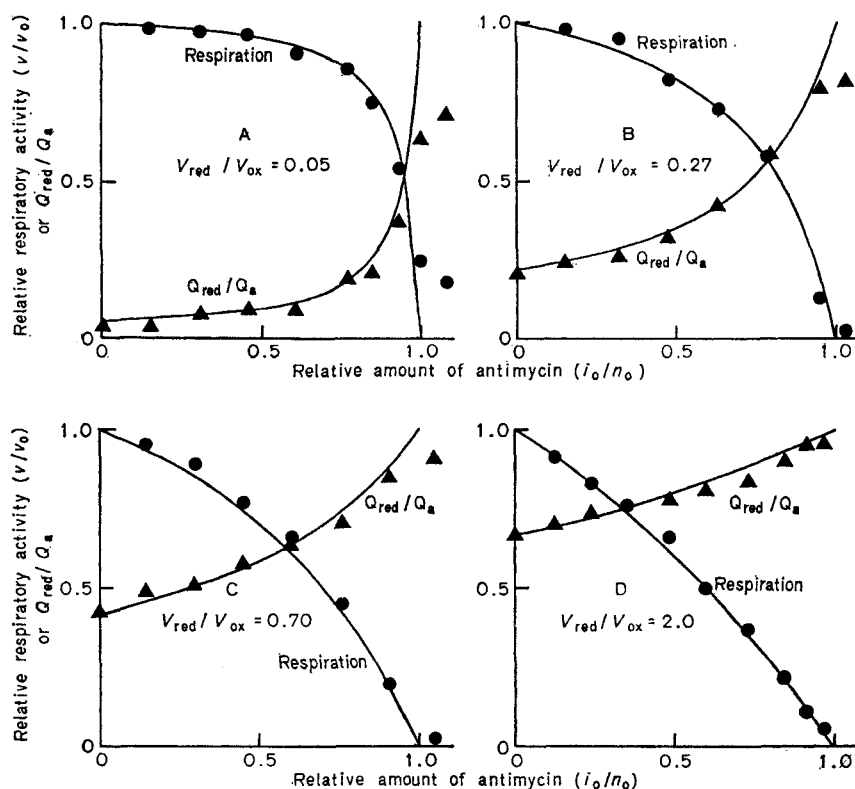


Fig.8. Antimycin titration of the respiratory activity and the degree of reduction of Q_a at various ratios V_{red}/V_{ox} . The respiratory activity and the degree of reduction of Q_a in the corresponding steady state of submitochondrial particles incubated in standard incubation medium at 25 °C with various concentrations of antimycin were determined as described under Methods. The relative respiratory activity (v/v_0) (●) and the degree of reduction of Q_a (Q_{red}/Q_a) (▲) are plotted against the amounts of antimycin (i_0) relative to n_0 . The curves are drawn according to Eqn (9) and Eqn (11) with

n/n_0 given by Eqn (5) and the given values of V_{red}/V_{ox} (from Table 1). Experimental conditions: A, normal submitochondrial particles (1.6 mg/ml) with NADH, 1.5 mM amytal and 5 μ M cytochrome *c*. In experiment B, C and D submitochondrial particles depleted from cytochrome *c* are used. (B) 0.63 mg/ml particles with 20 mM succinate, 0.2 mM malonate and 10 μ M cytochrome *c*. (C) 1 mg/ml particles with 20 mM succinate, 0.2 mM malonate and 0.5 μ M cytochrome *c*. (D) 0.7 mg/ml particles with NADH, 0.15 mM amytal and 0.1 μ M cytochrome *c*. Further details are listed in Table 1

The theoretical curves deviate from the experimental points near $i_0/n_0 = 1$. Whereas full inhibition of the respiration at this point is predicted by the model, a significant activity is found experimentally. Since the respiration is further decreased on increasing i_0/n_0 beyond unity, this discrepancy is considered to reflect the neglected dissociation of the inhibitor from the site as discussed in the following chapter.

The Influence of the Dissociation of the Inhibitory Complex on the Shape of the Titration Curve

The dissociation of the inhibitor complex (NI) according to Reaction (a) to give the free inhibitor site (N) and the free inhibitor (I) is accounted for by substituting Eqn (15) for Eqn (5).



Eqn (15) gives the relative amount of free inhibition sites (n/n_0) as a function of the relative amount

of the inhibitor (i_0/n_0) and of the dissociation constant (K_d) of Reaction (a) and results from the combination of Eqn (12) which describes the equilibrium of Reaction (a), with the conservation equations (13) and (14).

$$K_d = N \cdot I / NI \quad (12)$$

$$I_0 = I + NI \quad (13)$$

$$N_0 = N + NI \quad (14)$$

$$n/n_0 = 1/2[1 - K_d/N_0 - i_0/n_0 \pm \sqrt{(1 - K_d/N_0 - i_0/n_0)^2 - 4 K_d/N_0}]. \quad (15)$$

I and NI designate the concentration of free antimycin (I) and of the inhibitor complex (NI), respectively. $N_0(I_0)$ represents the total concentration of the inhibition sites (inhibitor) and is obtained from $n_0(i_0)$ on multiplication by the particle concentration used. With $K_d = 0$, Eqn (15) reduces to Eqn (5).

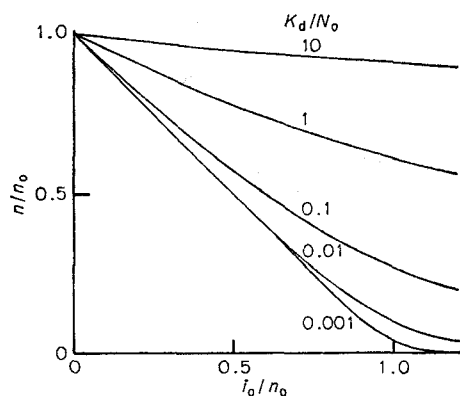


Fig. 9. Theoretical relation of the fraction of free inhibition sites (n/n_0) to the relative amount of the inhibitor (i_0/n_0) according to Eqn (15) with various values for K_d/N_0 .

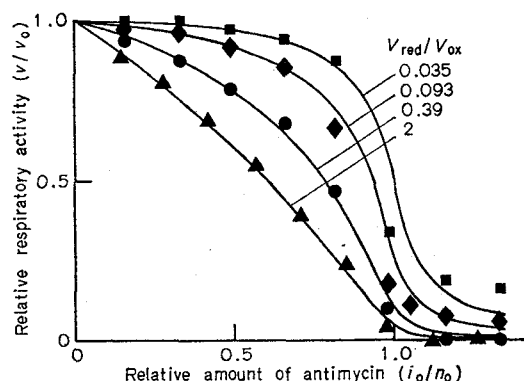


Fig. 10. Antimycin-titration of the respiratory activity at various donor activities and comparison to the theoretical curves implying K_d . The respiratory activity of a single preparation of normal submitochondrial particles (0.3 mg/ml) in standard incubation medium at 25 °C was titrated with antimycin at various donor activities as described under Methods. The relative respiratory activity (v/v_0) is plotted against the amount of antimycin (i_0) relative to n_0 (0.33 μ mol/g protein). The respiratory activity in the absence of antimycin (v_0) was (μ atoms $O \times \text{min}^{-1} \times \text{g protein}^{-1}$): (▲) 1220 with NADH, (●) 515 with succinate, (◆) 155 with 20 mM succinate and 1 mM malonate, (■) 62 with 20 mM succinate and 2 mM malonate. The curves are drawn according to Eqn (9) with n/n_0 given by Eqn (15) and $K_d/N_0 = 0.001$. The values given for V_{red}/V_{ox} were calculated from v_0 and V_{ox} according to Eqn (7). V_{ox} (1830 μ atoms $O \times \text{min}^{-1} \times \text{g protein}^{-1}$) was measured as the respiratory activity with durohydroquinone [1].

In Fig. 9, n/n_0 is plotted as a function of i_0/n_0 according to Eqn (15) with values of K_d/N_0 ranging between 0.001 and 10. With the values for K_d given by Berden and Slater [8] (0.3 nM and 0.03 nM) and $N_0 = 30$ nM as an experimentally reasonable concentration of the inhibition sites, essentially linear curves result which differ only slightly from the straight line given by Eqn (5). Since n/n_0 is assumed to be equal to V'_{ox}/V_{ox} [Eqn (4)] both curves are in

agreement with the linear inhibition curve of the electron transport through the cytochrome segment of the respiratory chain.

In the experiment of Fig. 10 the theoretical titration curves for the substrate oxidation incorporating $K_d = 0.1$ nM are compared to the experimental points obtained by antimycin titration of the respiration at various ratios V_{red}/V_{ox} . A single-particle preparation is used with a constant acceptor activity (V_{ox}) and only V_{red} is varied. This is done, firstly, by using NADH and succinate as the substrates and secondly, by inhibiting the succinic dehydrogenase with malonate.

The theoretical curves are drawn according to Eqn (9) with n/n_0 given by Eqn (15). K_d/N_0 is assumed to be 0.001 corresponding to $K_d = 0.1$ nM as the mean of the two values given by Berden and Slater, and $N_0 = 0.1$ μ M as the concentration of the inhibition sites under the actual experimental conditions. The values of V_{red}/V_{ox} are calculated from V_{ox} which was directly measured, and v_0 according to Eqn (7).

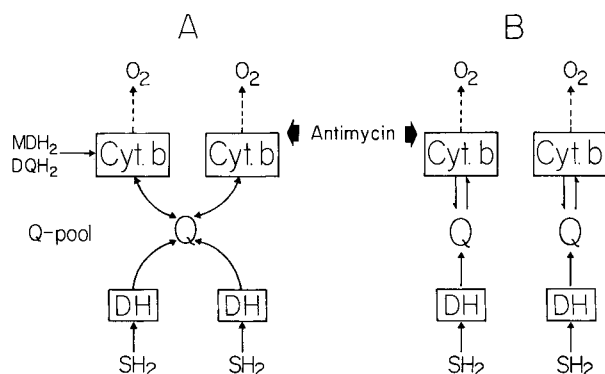
In agreement with the model, the relation of the respiratory activity to the antimycin concentration is the more hyperbolic the lower the donor activity. The theoretical curves are consistent in principle with the experimental points also in the region of equimolarity between antimycin and the inhibition sites ($i_0/n_0 = 1$). This indicates that this part of the inhibition curves is well accounted for by the introduction of K_d into Eqn (9). Especially the experimental fact that v/v_0 increases at $i_0/n_0 = 1$ with decreasing ratios V_{red}/V_{ox} , is in agreement with the theoretical curves. Also the order of magnitude of the K_d given by Berden and Slater [8] is confirmed.

DISCUSSION

The Pool Function of Q

As an obligatory respiratory component Q mediates the electron transfer serially between the dehydrogenase (donor) and the cytochrome section (acceptor) of the respiratory chain (Scheme 1). According to the results presented in this paper, Q also interacts as a "lateral" mediator between the cytochrome chains. Lateral as well as serial mediation of Q between the donor and the acceptor molecules (Scheme 1A) is required for the hyperbolic relation of the inhibition of substrate oxidation to the concentration of antimycin. An arrangement, however, with only serial mediation (Scheme 1B) would result in only linear relations to antimycin.

In particular the titration by antimycin of the respiratory activities with menadiol and durohydroquinone demonstrates that Q is responsible for the lateral mediation (Fig. 1–3). Although menadiol donates electrons at cytochrome *b* (Scheme 1), it can also reduce Q due to its sufficiently low redox poten-



Scheme 1. *Q* as a mediator of the electron transport in the respiratory chain. (A) Lateral and serial mediation, (B) serial mediation only. DQH₂, durohydroquinone; MDH₂, menadiol

tial by equilibration between cytochrome *b* and *Q* which is reversed to the "normal" direction of electron transport. Therefore, a hyperbolic antimycin titration curve is obtained with menadiol. When *Q* is missing after extraction or when *Q* cannot be reduced with a donor to cytochrome *b* of a too high redox potential such as durohydroquinone, the electrons flow through the cytochromes only, without the lateral mediation by *Q* resulting in a linear titration curve.

The lateral interconnection of respiratory chains has been disregarded by Potter and Reif [22] who originally proposed a kinetic explanation for the shape of the antimycin titration curve. Thorn [23] first discussed an interconnection by a diffusible electron carrier before the discovery of *Q*. Both authors silently implied a linear response to antimycin of the electron flow through the cytochrome chain which later was experimentally demonstrated by titration of the *Q*_{red}-cytochrome *c* reductase of complex III by Rieske *et al.* [24]. It should be mentioned that also for plastoquinone the role of a lateral mediator between the two photosystems of chloroplasts has been demonstrated [25].

The simplest mechanism feasible for the interconnection is the diffusion of *Q* (see [1–3, 6, 11, 13, 26]). This view is supported by the finding that the redox state of the total active pool of *Q* is a function only of the activity of the donor and acceptor segment of the respiratory chain, although *Q* is present in a considerable molar excess over the components which reduce and oxidize it. As a result, the redox response of *Q* is exactly that of a hydrogen-transporting mediator between the donor and the acceptor segment of the respiratory chain as defined by Eqn (10). This was demonstrated already in the preceding paper [1] by relating the respiratory activity to the redox state of *Q*, and is substantiated here by its response to the titration with antimycin

according to Eqn (11). Further support to the diffusional pool function is given by the fact that the redox active *Q* pool can be enlarged by the incorporation of *Q* into the membrane [1]. The inactive part of the content of *Q* (*Q*_i) is probably localized in particles which contain inactive dehydrogenases. This is suggested by the observation that the amount of *Q*_i in particles deficient of succinate dehydrogenase was decreased by about 40% on reinsertion of the dehydrogenase [2].

The Interaction of Antimycin with the Electron Transport

The kinetic model proposed for explaining the hyperbolic inhibition curve of substrate oxidation is based on the assumption that the electron transport through the cytochrome segment is inhibited proportional to the amount of antimycin bound. This assumption is confirmed by the experimentally demonstrated linear inhibition of the electron transport through the cytochrome segment (Fig. 1, 2, 4, 5) and by the finding that antimycin is extremely tightly bound [8]. Further support comes from the equality between the titer of antimycin inhibition and the content of cytochrome *b*_T [27–29] and from the linear inhibition of the ATP-induced reduction of cytochrome *b*_T by antimycin [29]. Therefore, it is likely that all the antimycin bound forms the inhibitory complex. In contrast, Berden and Slater [8] propose that the antimycin complex occurs in two states of which only one is inhibitory. The *K*_d of the inhibitory complex is postulated to be identical with that measured for the binding of antimycin to the oxidized particles (0.03 nM) and the *K*_d of the uninhibited complex with that measured for the reduced particles (0.3 nM). Antimycin is assumed to shift the equilibrium between the two states towards the inhibitory state, although the cooperativity of the binding is not significant. From the degree of inhibition of substrate oxidation, which is postulated to reflect the fraction of the complex in the inhibitory state, the allosteric parameters *n* and *L* are calculated. Since these values have not been tested by other independent methods, however, the hypothesis of allosteric interaction of antimycin cannot be considered to be experimentally founded.

The results given in this paper contradict the view that the interaction of antimycin with the respiratory chain causes the "nonlinear" inhibition curve of substrate oxidation. This is especially evident from the antimycin titration curve of the electron transport through the cytochrome chain in comparison to that of the respiratory activity of the same suspension of submitochondrial particles (Fig. 1 and 4). A strictly linear response of the electron flow through the cytochrome chain, as measured by the rate of oxidation of *Q*_{red} or as the respiratory activity with

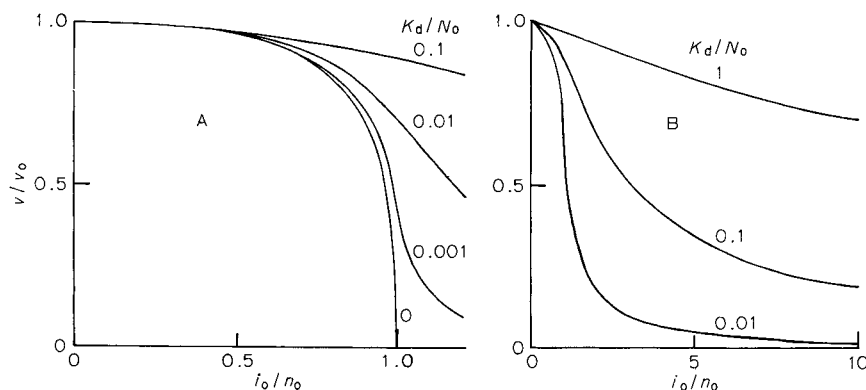


Fig. 11. Theoretical titration curves of the respiratory activity at various values for K_d . Relation of the relative respiratory activity (v/v_0) to i_0/n_0 according to Eqn (11) with $V_{\text{red}}/V_{\text{ox}} = 0.05$, n/n_0 given by Eqn (15) and various values for K_d/N_0 .

durohydroquinone or dimethoxy-dimethylbenzo-hydroquinone, is found together with a strongly hyperbolic titration curve of the respiratory activity with NADH and succinate. If the mode of action of antimycin were responsible for the shape of the curve, the same shape must be expected for the titration curves of both activities.

According to the "allosteric" hypothesis the variation of dehydrogenase activities should not influence the titration curve of the respiratory activity. However, in agreement with the kinetic model the curves become more hyperbolic with lower donor activities (Fig. 10). This is also confirmed with menadiol as the donor on variation of the donor activity by the concentration of D,T-diaphorase [5].

The finding that anionic detergents convert the hyperbolic titration curve of the succinate-cytochrome *c* reductase to a linear one, has been used as an argument in support of the hypothesis of "allosteric" inhibition [9]. A more convincing explanation on the basis of the kinetic model was recently published by Rieske and Gupta [14]. These authors demonstrated, that deoxycholate causes a drastic decrease of the activity of the Q_{red} -cytochrome *c* reductase at a concentration which does not affect the succinic dehydrogenase. Thus the ratio donor to acceptor activity increases considerably and a linear titration curve is expected according to the kinetic model.

The Relevance of the Kinetic Model for Other Inhibitors

In principle Eqn (9) is also relevant for other respiratory inhibitors either of the donor or the acceptor of Q , provided that they interact according to Eqn (5).

Rotenone and piericidin which inhibit at very low concentrations and are firmly bound to the respiratory chain, also seem to give titration curves

varying from linear to distinctly hyperbolic [30,31]. However, it is unclear so far, whether Eqn(9) can also be applied with these inhibitors, since their mechanism of action on the NADH-dehydrogenase segment is more complex [30].

The more linear inhibition curves observed with inhibitors like HQNO [32] and KCN can be explained on the basis of Eqn (9) with n/n_0 given by Eqn (15), if greater values for K_d are assumed. This is shown by Fig. 11, in which theoretical curves are drawn according to Eqn (9) and Eqn (15) with $V_{\text{red}}/V_{\text{ox}} = 0.05$ and various values for K_d/N_0 . The strongly hyperbolic shape of the curves found with smaller K_d/N_0 is obscured at greater values. Thus with $K_d/N_0 = 0.1$ appreciable inhibition is found only at $i_0/n_0 > 1$ (Fig. 11B) where the shape of the curve is mainly influenced by Eqn (15). This curve resembles that found with HQNO [32], which probably attacks the same site as antimycin, but is required in about ten-times greater concentration. A similar titration curve which may also be explained by a greater K_d was found with mitochondria of an antimycin-insensitive yeast mutant [33].

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